50-DAY DATA ANALYST CHALLENGE · EXCEL DAY 02

Banking & Finance – Loan Portfolio Analysis

Business situation

A bank wants to understand the loan portfolio: total principal disbursed, repayments received, outstanding balances and the contribution of each account to the portfolio. Analyse the sample accounts and prepare a concise report.

Tasks

1. Count the loan accounts and fields. Explain what one row represents and classify the fields by data type.

2. Keep Loan_Data unchanged. Create Loan_Analysis and convert the source range into an Excel Table named LoanPortfolio.

3. Add Outstanding_Balance = Loan_Amount − Amount_Repaid.

4. Add Repayment_Percentage = Amount_Repaid ÷ Loan_Amount and format as percentage.

5. Add Portfolio_Share = each Loan_Amount ÷ total Loan_Amount. Use an absolute reference for the total and copy the formula down.

6. In an Assumptions sheet, enter a hypothetical annual simple-interest benchmark of 10.5%. Calculate an illustrative annual interest amount on each original principal using an absolute reference. Clearly label this as an estimate, not actual contractual interest.

7. Use SUM, AVERAGE, MIN, MAX, COUNT/COUNTA and ROUND to summarize total loan, total repaid, total outstanding, average loan, smallest loan and largest loan.

8. Compare outstanding balances by region and loan type. Identify the largest outstanding individual account.

9. Write two evidence-based observations, one recommendation and one limitation of the data.

Reference challenge

Demonstrate a relative reference and an absolute reference. Explain why dollar signs are needed in an absolute reference. Mixed references ($A1 and A$1) are an extension task: create a small two-way grid and explain which part remains fixed.

Submission

Submit Day02_Loan_Analysis.xlsx with Loan_Data (unchanged), Loan_Analysis, Assumptions and Business_Insights worksheets, plus README.md containing your method, key findings, recommendation and assumptions.

GitHub structure

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50-Day-Data-Analyst-Challenge/Excel/Day-02/Day02_Loan_Analysis.xlsx
50-Day-Data-Analyst-Challenge/Excel/Day-02/README.md
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Integrity: Work independently. Do not hard-code calculated values. The 10.5% benchmark is an exercise assumption only, not a real bank rate.